Smart Wattanapornmongkol

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Profile

Engineering student at Chulalongkorn University working across data-centric machine learning, Thai ASR, medical imaging, backend systems, and academic program operations. Current work includes AI Academy operations at the Artificial Intelligence Association of Thailand, student research assistance on monocular colonoscopy polyp size estimation, backend leadership at CU NEX, and full-stack development for BIZCUBE Chula. Research output includes an IEEE ICCI 2024 publication on contextual matching between music and image.

Research Interests

Data-centric machine learning; low-resource ASR; synthetic speech and data selection; Thai language AI; structure from motion; physics-informed neural networks; backend infrastructure for education and student systems.

Education

B.Eng. in Information and Communication Engineering — Aug 2024 - Oct 2028 expected
Chulalongkorn University, International School of Engineering — Bangkok, Thailand

- CGPA: 3.88/4.00.
- ISE50 Academic Excellence Scholarship, academic year 2025.

High School Diploma, EPLUS+ Program — May 2018 - Mar 2024
Suankularb Wittayalai School — Bangkok, Thailand

- Completed junior-senior high school in the EPLUS+ Program.

Current Appointments & Experience

Academy Secretary — Jun 2026 - Present
Artificial Intelligence Association of Thailand — Thailand

- Coordinate AI Academy program operations and teaching delivery for Super AI Engineer Season 6: Digital Workforce Transformation and partner curricula.
- Coordinate teaching logistics, mentor communication, learner follow-up, and project tracking for AI training activities.
- Prepare session materials and consolidate program issues for instructors and program leads.

Student Research Assistant — Mar 2026 - Present
Chulalongkorn University — Bangkok, Thailand

- Work on structure-from-motion methods for monocular colonoscopy polyp size estimation using physics-informed neural networks.
- Support geometric reconstruction and medical-image measurement research while continuing undergraduate study in Information and Communication Engineering.

Head of IT & Backend Lead; Back End Developer — Jul 2024 - Present
CU NEX Club, with Kasikorn Business-Technology Group — Bangkok, Thailand

- Lead IT and backend engineering for CU NEX and coordinate development with Kasikorn Business-Technology Group and university stakeholders.
- Architected and delivered election-management tools, including candidate/admin panels and candidate forms, used by 27+ admins and 255+ candidates.
- Defined requirements and acceptance criteria for the CU NEX attendance-check feature; coordinated backend APIs and reporting to reduce manual reconciliation by 60%.
- Established code review, CI/CD, automated tests, and deployment procedures for backend releases.

- Manage and mentor a cross-functional team of 12 developers through sprint planning, prioritization, and stakeholder communication.

Technical Lead; Back End Developer; Inner of Secretary Department Jul 2024 - Present
International School of Engineering Student Committee, Chulalongkorn University Bangkok, Thailand

- Serve as Technical Lead from Jun 2026, following Back End Developer work from Jul 2025 to Jun 2026 and Inner of Secretary Department service from Jul 2024 to Jul 2025.
- Coordinate student-committee technical work and maintain internal systems used by the International School of Engineering community.

Fullstack Developer Mar 2026 - Jul 2026
BIZCUBE Chula, Chulalongkorn Business Administration Bangkok, Thailand

- Implemented course-selling and payment interfaces handling approximately 8M baht in total transactions.
- Built user-facing purchase flows and administrative workflows for course commerce.
- Supported operations and marketing teams with data and course logistics, using agentic workflows to improve task productivity by 50%.

AI Researcher Intern, Speech & Multimodal Learning May 2025 - Mar 2026
OpenThaiGPT Lab / iApp Technology / Super AI Engineer SS5 Research Track Bangkok, Thailand

- Led a data-centric study on synthetic speech selection for low-resource Thai ASR, moving from naive scaling to quality-aware sample selection for Whisper fine-tuning.
- Proposed Audio-Conditioned Gain, a normalized metric for sample utility based on marginal gains over a text-only baseline.
- Used the metric to identify informative synthetic samples without additional annotation.
- Showed bottom-k Audio-Conditioned Gain selection outperformed random and top-k baselines across CV, FLEURS, and Porjai evaluations.

Publications

Direct matching between music and image for contextual relationship analysis 2024
IEEE ICCI 2024. DOI: 10.1109/ICCI60780.2024.10532575.

Teaching, Mentoring & Service

Teaching Team Aug 2026 - Sep 2026
AIAT x College of Arts, Media and Technology, Chiang Mai University AI Entrepreneur Curriculum Chiang Mai, Thailand

- Teach five sessions: an LLM prompting and Gemini Notebook foundation lecture; labs on RAG, security for AI applications, agentic AI applications, and deploying AI-enabled systems.

Teaching Assistant, ITE68367 Special Topics in Digital Technology 2 Jun 2026 - Jul 2026
Master of Science in Information Technology, Sripathum University Thailand

- Develop the course LMS, grade assignments, and teach a session on agentic AI tools and LLM prompting.

Teaching Assistant, Super AI Engineer Season 6 May 2026 - Jun 2026
Artificial Intelligence Association of Thailand Pathum Thani, Thailand

- Assisted instructors during lectures and supported learning activities for Digital Workforce Transformation.
- Set up Kaggle-based competition workflows and consolidated hackathon scores for program review.
- Shared applied AI project experience with participants during program sessions.

Teaching Assistant 21-24, 27 Apr 2026
Royal Thai Air Force Thailand

- Teach one session on Graph RAG and assist the lecturer during other sessions.

Coach Feb 2026 - Apr 2026
National Electronics and Computer Technology Center (NECTEC) Bangkok, Thailand

- Coached participants in technical learning activities connected to AI and engineering practice.

Selected Systems & Projects

Election Management System (EMS) Aug 2024 - Jun 2025
Internal platform covering candidate/admin panels, candidate forms, backend APIs, reporting, tests, and deployment practices. Used by 27+ admins and 255+ candidates.

QuickAttend Jul 2025 - Jun 2026
Web-based attendance platform that lets students and activity staff scan CU NEX QR codes to register participation, with centralized attendance records for activity organizers.

BIZCUBE Chula course commerce and payment interface Mar 2026 - Jul 2026
Course-selling and payment workflow for Chulalongkorn Business Administration activities, handling approximately 8M baht in total transactions.

Audio-Conditioned Gain research code and experiments 2025 - 2026
Research workflow for low-resource Thai ASR sample selection, comparing bottom-k, random, and top-k strategies across CV, FLEURS, and Porjai evaluation sets.

MySK school platform 2022 - 2024
School digital transformation project for Suankularb Wittayalai students and teachers. Provided unified access to school resources and information; used by 5,000+ students and 200+ teachers and directors.

Honors

- ISE50 Academic Excellence Scholarship, academic year 2025, Chulalongkorn University.

Technical Skills

Languages	Python, TypeScript, SQL, C/C++, Rust
AI/ML	Pandas, PyTorch, Keras, HuggingFace Transformers, Whisper, W&B
Backend	FastAPI, Actix Web, Node.js, REST APIs, PostgreSQL
Tools	Git, Docker, CI/CD, Linux, AWS, GCP, Azure